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Multi-Modal Deepfake Detection: Integrating Spatial, Temporal, and Physiological Features

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1 Introduction

Deepfake technology refers to AI-created content where a person’s facial appearance has been artificially modified or replaced. With the current state-of-the-art generative models, particularly Generative Adversarial Networks (GANs)^[1] and autoencoders, it is now possible to produce manipulated videos and images that are almost indistinguishable from real media^[2]. Though such innovations offer beneficial applications in entertainment and accessibility, their abuse presents profound societal and technical challenges. Misuse comes with a high cost in areas like misinformation operations, identity theft, political propaganda, and cyber-fraud^[3].

The proliferation of hyper-realistic synthetic media threatens to erode public trust, destabilize political discourse, and enable new forms of personal harassment. Successive high-profile cases involving spurious orations and doctored videos of celebrities have highlighted the urgent need for reliable detection methods. As the realism of this artificially created material increases, traditional forensic indicators like obvious editing or compression artifacts are no longer reliable, making sophisticated and automated detection methods a critical requirement^[4]. This project addresses this need by proposing a novel multi-modal detection framework that combines different analytical approaches to create a more robust and resilient system.

2 Motivation

Notwithstanding recent developments, current deepfake detection methods are subject to several critical shortcomings that limit their real-world effectiveness. Each dominant paradigm, when used in isolation, has a distinct failure mode. For instance, a CNN-based model meticulously trained on GAN-generated fakes may be easily bypassed by newer synthesis techniques like diffusion models. Temporal models, which analyze video sequences, could be fooled by a visually perfect deepfake that simply lacks biological authenticity—an artifact they are not designed to detect. Finally, promising physiological methods like rPPG can be too fragile for practical use, as a short, compressed clip from social media might not provide enough clean data for a conclusive analysis.

To address these individual issues, this project is motivated by the need for a holistic solution that is greater than the sum of its parts. We introduce a multi-modal deepfake detection system that integrates complementary modalities to create a layered defense:

- A **spatiotemporal feature extractor (ResNeXt-50 CNN + LSTM)** for detecting visual and sequential artifacts.
- An **rPPG-based physiological module** for detecting the absence of authentic biological signals.

By intelligently fusing evidence from spatial, temporal, and physiological streams, the proposed system aims to achieve greater accuracy, generalizability, and robustness. This approach offers a stronger, more resilient defense against the constantly evolving threat of deepfakes.

3 Literature Survey

The rapid evolution of deepfake technology has spurred a corresponding surge in research dedicated to its detection. These methods can be systematically divided into three major directions:

1. **CNN-Based Spatial Detection:** Convolutional Neural Networks (CNNs) are used extensively to detect spatial-level artifacts present in forged video frames. These models are trained to spot inconsistencies that are often invisible to the naked eye, such as unnatural reflections in the eyes, "checkerboard" artifacts from up-sampling layers, or microscopic imperfections in facial textures. Architectures like XceptionNet^[5], ResNet^[6], and DenseNet^[7] have shown success in these areas. However, their primary limitation is a lack of generalization; they often overfit to the specific artifacts of the datasets they were trained on and their performance degrades significantly with common post-processing like video compression^[8].
2. **Temporal and Multi-Modal Methods:** To overcome the limitations of static frame-level analysis, researchers have explored temporal modeling using hybrid CNN-RNN architectures, particularly those with Long Short-Term Memory (LSTM) networks^[9]. These systems can identify temporal irregularities like unnatural eye-blinking rates^[10], inconsistent head movements^[11], or flawed lip-syncing. Furthermore, multi-modal systems that incorporate both visual and audio features have demonstrated superior performance over unimodal methods^[12], though they are subject to challenges related to data synchronization and computational scalability.
3. **Physiological Signal Analysis:** One of the most recent frontiers involves analyzing physiological signals that are exceptionally difficult for generative models to reproduce accurately. Remote photoplethysmography (rPPG)^[13], for example, infers heart-rate data from minute color changes in facial regions caused by blood flow^[14]. Faking this signal would require not just a perfect visual rendering but a simulation of the human cardiovascular system's response to speech, emotion, and movement—a task of immense complexity. Because these biological signals are hard to forge persuasively, they serve as strong evidence of a video's authenticity. Research in this area has shown robustness against demographic changes and adversarial attacks, though performance can be hampered by environmental noise and low-quality video^[15].

4 Problem Statement

The core problem this research addresses is the growing gap between the sophistication of deepfake generation and the effectiveness of current detection methods. This gap is defined by the specific technical failures of existing detection paradigms when applied in real-world scenarios.

Current systems, which typically rely on a single modality, have demonstrated the following critical weaknesses:

- **Lack of Generalization:** Visual detectors (CNNs) often fail when confronted with new, unseen deepfake generation methods.
- **Vulnerability to High-Quality Fakes:** Temporal models (CNN-LSTMs) can be deceived by visually flawless fakes that lack underlying biological authenticity.
- **Fragility in Noisy Conditions:** Physiological methods (rPPG) are often too sensitive to video quality, compression, and lighting to be reliable on their own.

Considering these individual failures, the central problem is the lack of a robust, multi-faceted detection framework that overcomes these isolated weaknesses. There is a critical need

for an integrated system that can intelligently fuse evidence from multiple, complementary domains—spatial, temporal, and physiological—to create a detector that is not only accurate but also generalizable and resilient to the evolving complexity of synthetic media^[16].

5 Objective and Methodology

The primary objective of this project is to design, implement, and evaluate a robust multi-modal deepfake detection system that achieves high accuracy by integrating spatio-temporal visual analysis with physiological signal authentication.

Secondary objectives include:

- To develop a novel physiological credibility score for quantifying the authenticity of an rPPG signal.
- To create an adaptive fusion algorithm that dynamically weights evidence from visual and physiological streams based on confidence.
- To evaluate the proposed system’s performance against state-of-the-art unimodal detectors on benchmark datasets.

5.1 System Overview

Our deepfake detector uses a multi-modal architecture to capture a wide array of forgery artifacts. As shown in Figure 1, the framework comprises two parallel analysis streams: a CNN-RNN hybrid model for spatio-temporal feature extraction and an rPPG-based module for physiological signal analysis. The outputs from these streams are integrated via an intelligent fusion system to produce a final classification.

5.2 Model Architecture

The system’s architecture is built upon three specialized modules designed to work in concert: a spatial analysis module to inspect individual frames, a temporal module to analyze sequences, and a physiological module to detect biological signals. This multi-pronged approach ensures a comprehensive analysis of the input video.

5.2.1 Spatial Analysis Module (CNN)

The foundation of our visual analysis stream is a pre-trained ResNeXt-50 Convolutional Neural Network (CNN). This module processes individual video frames to identify tell-tale spatial anomalies that are often hallmarks of digital manipulation, such as blending artifacts at the edge of the swapped face, unnaturally smooth skin textures, and inconsistent lighting or shadows. For each frame, the ResNeXt-50 outputs a 2048-dimensional feature vector.

5.2.2 Temporal Analysis Module (RNN)

The sequence of feature vectors generated by the spatial module is then fed into a Long Short-Term Memory (LSTM) network. The LSTM’s role is to analyze the video’s temporal dynamics, looking for inconsistencies that emerge over time, such as aberrant eye-blinking patterns or unnatural facial transitions. The integrated spatio-temporal architecture is illustrated in Figure 2.

Multi-Modal Deepfake Detection Architecture

Integrating Spatial, Temporal, and Physiological Analysis

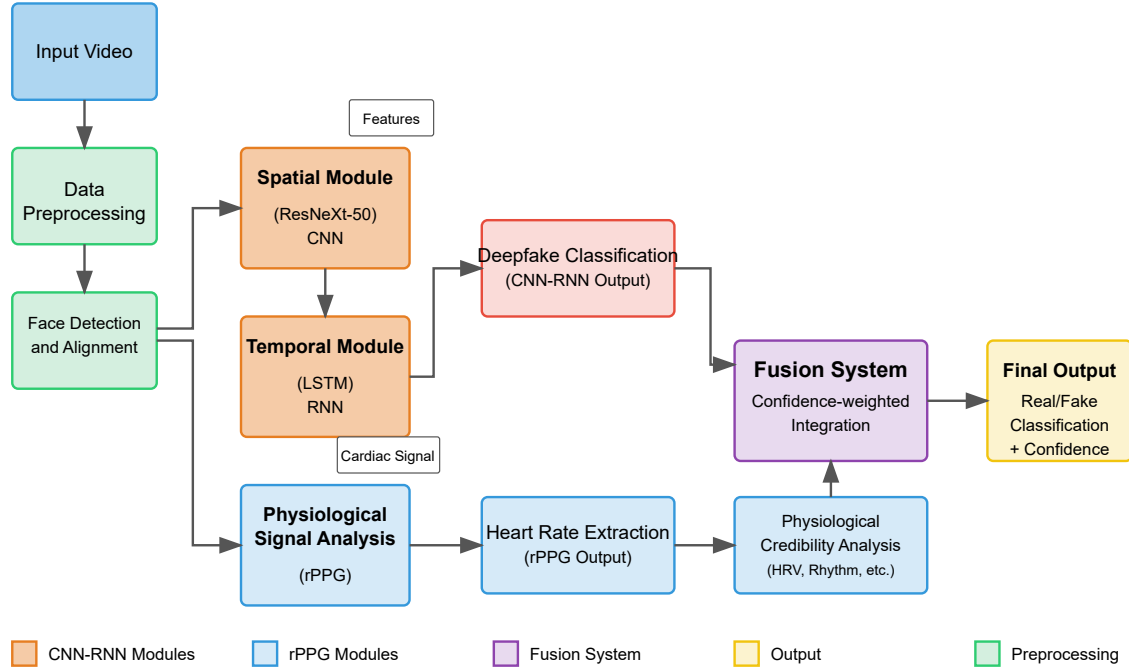


Figure 1: High-level architecture of the proposed multi-modal deepfake detection system, integrating visual and physiological analysis.

CNN-RNN Model Architecture

Spatio-Temporal Detection



Figure 2: The integrated ResNeXt-50 and LSTM architecture for spatio-temporal feature extraction.

5.2.3 Physiological Signal Analysis Module (rPPG)

Operating in parallel, the physiological module uses remote photoplethysmography (rPPG) to measure heart rate from the video by monitoring minute chromatic changes in facial skin regions. The presence of a stable and biologically plausible heart signal is strong evidence of authenticity. This module provides an orthogonal signal of authenticity—one that is independent of visual artifacts—making it a powerful tool against even the most visually convincing deepfakes, as illustrated in Figure 3.

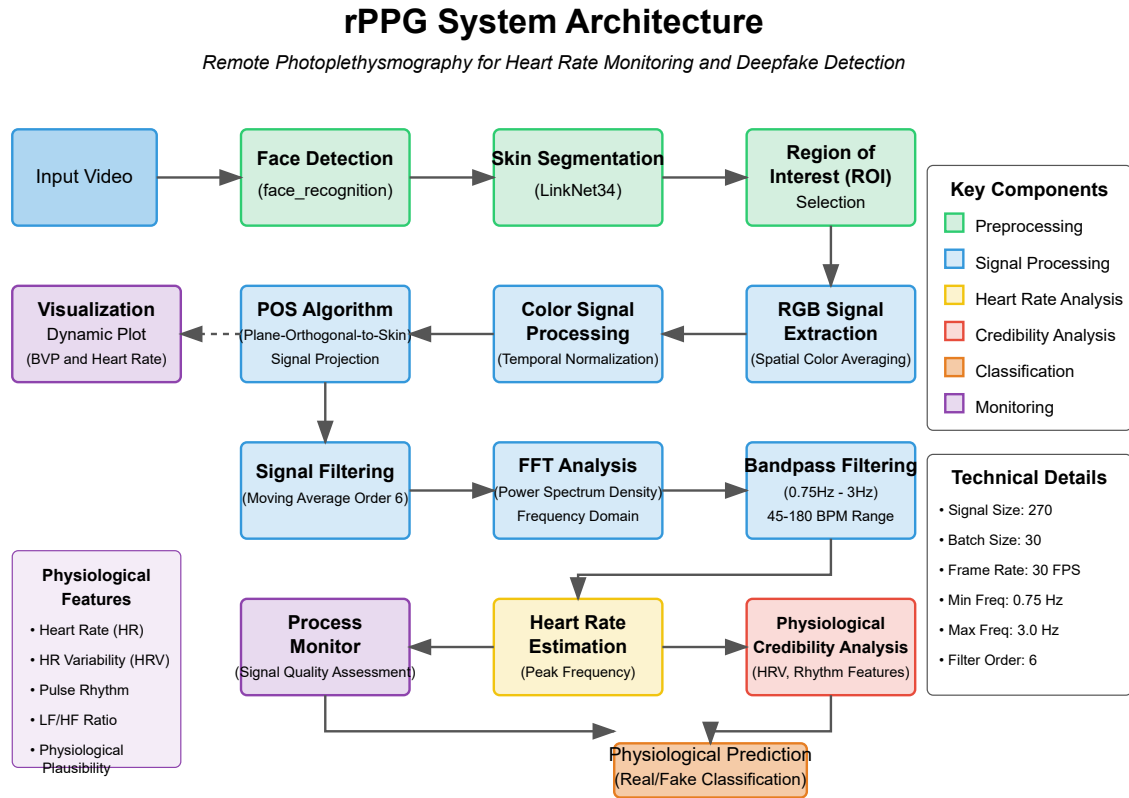


Figure 3: Architecture of the rPPG-based physiological signal analysis module.

5.3 Multi-Modal Fusion Methodology

To fuse the outputs from the visual (CNN-RNN) and physiological (rPPG) streams, we designed a confidence-weighted fusion algorithm. The logic relies on adaptive weighting, physiological plausibility scoring, and systematic handling of model agreement and disagreement.

A key component is the **Physiological Credibility Score** (Φ), a metric from 0-100 that quantifies the biological validity of the rPPG signal based on the criteria in Table 1.

Table 1: Physiological Credibility Scoring Criteria

Metric	Condition	Score
Mean Heart Rate	$65 \leq \mu(HR) \leq 120$ BPM	+20
HRV (RMSSD)	$5 \leq \text{RMSSD} \leq 50$ ms	+20
	$0.25 \leq \text{RMSSD} < 5$ or $50 < \text{RMSSD} \leq 80$ ms	+10
HRV (pNN50)	$\text{pNN50} \geq 5\%$	+20
	$2\% \leq \text{pNN50} < 5\%$	+10
Frequency (LF/HF Ratio)	$0.5 \leq \text{LF/HF} \leq 2.0$	+20
	$0 \leq \text{LF/HF} < 0.5$ or $2.0 < \text{LF/HF} \leq 3.0$	+10
Signal Stability	$1.5 \leq \sigma(HR) \leq 15$	+20
	$0.5 \leq \sigma(HR) < 1.5$ or $15 < \sigma(HR) \leq 25$	+10
Total Score		Max: 100

The complete fusion logic is detailed in Algorithm 1. It includes a fallback condition for unreliable rPPG signals, adaptive weight calculations based on model confidences, and specific rules for handling cases where the models agree (boosting confidence) or disagree (using an override or penalty system).

Algorithm 1: Multi-Modal Fusion Logic

Input: P_V, C_V, P_P, C_P , Heart Rate data HR **Output:** Final Prediction P_F , Final Confidence C_F

```
1 if  $|HR| < 4$  or  $C_P < 10$  then
2   return  $P_V, C_V$  // Fallback to visual model
3 else
4    $\Phi \leftarrow \text{CalculatePhysiologicalCredibility}(HR)$ ;
5    $w_V, w_P \leftarrow \text{CalculateAdaptiveWeights}(C_V, C_P)$ ;
6   if  $C_P > 20$  and  $\Phi > 0$  then
7      $\hat{C}_P \leftarrow \Phi$ ;
8   else
9      $\hat{C}_P \leftarrow C_P$ ;
10  if  $P_V = P_P$  then
11    // Case A: Model Agreement
12     $P_F \leftarrow P_V$ ;
13     $C_F \leftarrow \min(100, (C_V * w_V + \hat{C}_P * w_P) * 1.3)$ ;
14  else
15    // Case B: Model Disagreement
16    if  $(\hat{C}_P * w_P > C_V * w_V * 1.5)$  and  $(C_P > 30)$  then
17       $P_F \leftarrow P_P$ ;
18      // rPPG Override
19       $C_F \leftarrow (\hat{C}_P * w_P) * 0.8$ ;
20    else
21       $P_F \leftarrow P_V$ ;
22      // CNN-RNN Primary
23       $C_F \leftarrow (C_V * w_V) * 0.8$ ;
24  return  $P_F, C_F$ ;
```

6 Simulation Platform and Requirements

6.1 Simulation Platform

- **Hardware:** The model training and evaluation were conducted on a system equipped with an NVIDIA GeForce RTX 3060 GPU (12GB VRAM), an Intel Core i7-12700H CPU, and 16GB of RAM.
- **Software:** The primary development environment was Python 3.9. Key libraries included PyTorch (v1.12) for deep learning, OpenCV (v4.5) for video processing, the `face_recognition` library for facial detection, and Scikit-learn for performance metric calculations.

6.2 Dataset and Preprocessing

The models were trained and evaluated on public benchmark datasets, including FaceForensics++^[17] and the Deepfake Detection Challenge Dataset (DFDC). Preprocessing involved several crucial

stages:

1. **Frame Extraction:** Videos were sampled at 30 fps. Short sequences of 20 frames were used for the CNN-RNN model, while longer 270-frame sequences were used for rPPG analysis.
2. **Face Detection and Alignment:** Faces were detected using the `face_recognition` library and aligned using eye landmarks to normalize rotation and scale.
3. **Signal Extraction and Normalization:** A LinkNet34 network was used for skin segmentation, and a POS algorithm was applied to derive the raw blood volume pulse signal. Face frames were normalized using ImageNet’s statistics.

7 Results and Discussion

7.1 CNN-RNN Model Performance

The core CNN-RNN visual model demonstrated strong learning capacity and good generalization. As shown in Figure 4, the training and validation accuracy curves progressed smoothly, achieving a peak validation accuracy of 98.13% without significant evidence of overfitting. The corresponding loss curves in Figure 5 corroborate this, with the validation loss converging to a stable minimum of 0.048.

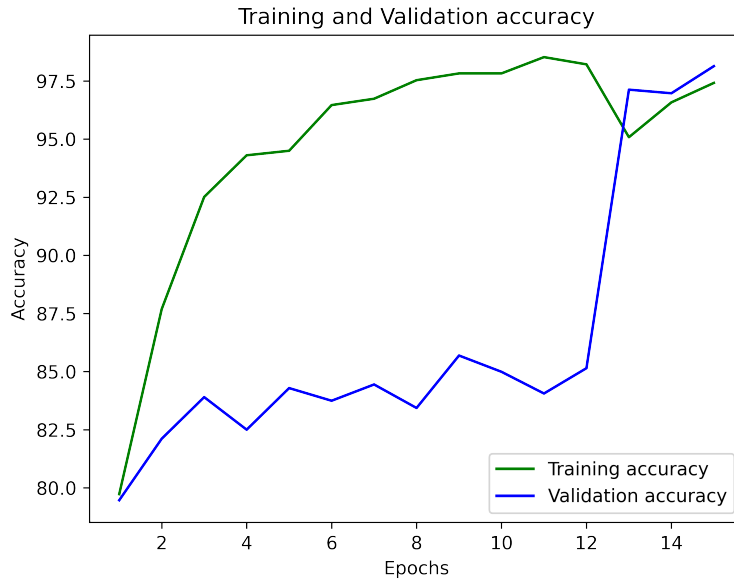


Figure 4: Training and validation accuracy for the CNN-RNN model over 20 epochs.



Figure 5: Training and validation loss for the CNN-RNN model over 20 epochs.

7.2 Fused System Performance on Unseen Data

To evaluate real-world effectiveness, the complete multi-modal system was tested on a held-out, unseen dataset. The fused system achieved an overall testing accuracy of **92.4%**. A detailed breakdown reveals the model correctly identified 460 fake samples (True Positives) and 470 real samples (True Negatives). The 34 false negatives (fakes classified as real) are the most critical errors in a security context, as they represent failures to detect a threat. The 42 false positives, while less critical, still highlight areas for improving model precision. These results underscore the effectiveness of the fusion approach but also point to specific areas for future enhancement.

7.3 Comparative Analysis

Our multi-modal framework achieves performance competitive with state-of-the-art methods, as shown in Table 2. While some models may achieve higher accuracy on specific datasets, our framework’s strength lies in its robustness, derived from physiological verification. This additional layer provides an orthogonal source of evidence, making it more resilient to highly realistic deepfakes that might fool purely visual systems. The superior performance, especially on the challenging DFDC dataset, highlights the tangible benefit of our fusion methodology.

Table 2: Performance comparison with other methods on the DFDC and Celeb-DF datasets.

Model	DFDC (Accuracy)	Celeb-DF (Accuracy)
XceptionNet ^[5]	63.2%	59.2%
ResNet-50 ^[6]	64.6%	67.1%
ResNet+LSTM ^[18]	69.0%	70.9%
ResNeXt+LSTM ^[19, 20]	90.9%	87.2%
Our Method (Fused)	92.4%	89.0%

8 Conclusion

This project successfully designed and implemented an effective multi-modal system for detecting deepfakes by jointly integrating spatio-temporal visual verification and physiological signal authentication. Our CNN-RNN hybrid model excelled at detecting visual anomalies, while the rPPG component supplied an auxiliary channel for recognizing hard-to-forge biological signals. The integrated system reached a final testing accuracy of 92.4

For future work, we propose several research directions:

1. **Dataset Expansion:** Training on more diverse datasets covering a broader range of video qualities, resolutions, and demographic profiles will enhance robustness.
2. **Vision Transformer (ViT) Integration:** Exploring the integration of Vision Transformers as an alternative spatial feature extractor. Unlike CNNs, which excel at local features, ViTs can capture global relationships across an image, potentially offering a more holistic analysis for detecting subtle, large-scale manipulation artifacts.
3. **Explainability (XAI):** Incorporating explainable AI methods like LIME or SHAP would build trust and allow for better failure analysis of the model's decisions^[21].
4. **Adversarial Training:** Using the detector as a discriminator within a GAN framework would allow it to continuously adapt to new deepfake generation techniques, improving long-term resilience.
5. **Scalability and Deployment:** Optimizing the system for large-scale, real-time cloud deployment is crucial for practical application while ensuring privacy and efficiency.

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